ChatGPT

Measurement Foundations for a Custom Browser-Based Web Performance Profiler

Executive synthesis

As of **29 August 2026**, the current stable **Core Web Vitals (CWV)** are **Largest Contentful Paint (LCP)** for loading performance, **Interaction to Next Paint (INP)** for responsiveness, and **Cumulative Layout Shift (CLS)** for visual stability. Google's current "good" thresholds are LCP $\leq$2.5 s, INP $\leq$200 ms, and CLS $\leq$0.1, normally assessed at the 75th percentile of page views. **First Input Delay (FID) ceased to be a Core Web Vital when INP replaced it in March 2024.** FCP, TBT, Speed Index, TTI, TTFB, memory measurements, network timings, and browser CPU/rendering counters are not Core Web Vitals. [1]

This distinction is more consequential than terminology suggests. Core Web Vitals were conceived primarily as **field-oriented user-experience outcome metrics**. LCP and CLS can also be measured meaningfully in controlled laboratory runs, while INP requires actual or scripted interactions; a navigation-only Lighthouse run therefore cannot directly measure INP and uses **Total Blocking Time (TBT)** as a laboratory diagnostic correlated with responsiveness rather than as an equivalent metric. Google explicitly warns that TBT is neither a Core Web Vital nor a field substitute for INP. [2]

A rigorous profiler should consequently distinguish at least three kinds of measurements. The first are **user-centric outcome indicators** such as LCP, INP, CLS, FCP, and, in synthetic testing, Speed Index. The second are **mechanistic or diagnostic measurements** such as request timings, resource sizes, long tasks, JavaScript execution, style recalculation, layout, and paint activity. The third are **resource/state measurements**, such as JavaScript heap size, page-attributed memory estimates, DOM-node counts, document counts, and event-listener counts. These classes answer fundamentally different questions and should not be collapsed into a single undifferentiated notion of "performance." W3C specifications themselves frame Paint Timing, Event Timing, Navigation Timing, Resource Timing, Long Tasks, and Server Timing as different observable phenomena rather than one global performance variable. [3]

A second major conclusion is that **a browser profiler does not directly observe several quantities that developer terminology appears to imply**. TTFB is not "server processing time": from a browser it is an end-to-end client-observed interval ending at `responseStart`, potentially incorporating redirects, cache/CDN behavior, connection establishment, and network transport. Server-internal processing can only be distinguished when the server explicitly supplies measurements, for example through `Server-Timing`. Similarly, Resource Timing's `transferSize` is not exact wire traffic: the specification deliberately uses a fixed 300-byte header approximation and special values for cache behavior. [4]

Likewise, there is no standards-defined universal metric called "JavaScript execution time," "rendering time," "layout time," or "painting time." Chrome DevTools presents categories such as Scripting, Rendering, and Painting by interpreting Chromium trace events; examples include Evaluate Script, Function Call, Recalculate Style, Layout, Composite Layers, and Paint. Those are extremely useful diagnostic signals, but they are

Chromium instrumentation semantics, not cross-browser Web Performance API metrics with the interoperability status of Navigation Timing or Resource Timing. [5]

The same caution applies to memory. `performance.memory` is non-standard and reports JavaScript-heap quantities whose scope can both under- and over-represent a page. The WICG `performance.measureUserAgentSpecificMemory()` proposal instead estimates memory attributable to a web application, including its workers and iframes, but requires cross-origin isolation and deliberately permits implementation-specific estimation and breakdown granularity. Chrome DevTools Protocol's `Runtime.getHeapUsage`, meanwhile, reports the JavaScript heap of the corresponding V8 isolate and explicitly states that it is not scoped to a particular runtime context. These measurements should therefore never be presented as interchangeable "page memory." [6]

Construct validity is an equally important limitation. LCP does not claim to measure whether the most *important* content is ready; its specification explicitly characterizes LCP as a heuristic and acknowledges that it can be wrong, including when early interaction terminates observation before the eventual main content appears. CLS quantifies layout instability, not usability in general. INP measures the latency of a defined set of interactions, not scrolling, animation smoothness, or application correctness. Speed Index quantifies visual progression, not functional readiness. [7]

Empirical systems and QoE research reinforces this limitation. Kelton et al. found that conventional loading metrics poorly represented user-perceived page-load time and showed that visual attention matters; their later mobile work found that initial-viewport models inadequately represent mobile users who begin scrolling while loading. Netravali et al.'s Vesper study reported that existing loading metrics could under- or overestimate their explicitly functional notion of time-to-interactivity by 24–64%. More recent controlled and crowdsourced work by Wehner and colleagues found Core Web Vitals substantially less predictive of subjective web QoE than might be inferred from treating them as comprehensive UX measures. [8]

The strongest measurement claim a custom profiler can therefore make is not "we measure web performance" in a unitary sense. It can legitimately claim that it **observes specified browser events and state, derives explicitly versioned metrics from them, and characterizes several distinct dimensions of runtime performance under declared execution conditions**. Any stronger claim—especially one equating a synthetic score with real-user experience, client-side TTFB with server execution, TBT with INP, trace CPU categories with cross-browser standards, or heap size with total page memory—requires additional qualification. [9]

Metric evidence matrix

The table treats a measurement as **direct** when the browser exposes the relevant event, timestamp, counter, or trace interval as the immediate observation, and as **derived** when a profiler must aggregate, select, subtract, threshold, window, or otherwise transform lower-level observations. That distinction is conceptual rather than metaphysical: even "direct" browser entries depend internally on browser algorithms. High Resolution Time further permits user agents to reduce clock precision for security and privacy, so timestamps should not be described as inherently nanosecond- or even sub-millisecond-accurate observations. [10]

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
First Contentful Paint (FCP)	Beginning of visible content presentation; time, normally ms	The first time the user agent paints content originating from the document rather than merely the initial/default background. It is reported as `PerformancePaintTiming` with a timestamp relative to the performance time origin. A document with no qualifying content may never generate FCP.	Paint Timing `first-contentful-paint` entry. **Direct browser milestone** under the specified paint algorithm.	Both
Largest Contentful Paint (LCP)	Perceived loading of prominent viewport content; time, ms	Browser emits candidates as qualifying content is rendered; the metric tracks the largest eligible contentful element seen in the viewport. Observation is terminated by defined user interaction/scroll conditions. `LargestContentfulPaint.startTime` is based on its render time.	LCP performance entries plus candidate-selection/ termination algorithm. **Derived selection over observed paints.**	Both; field is canonical for CWV

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
Cumulative Layout Shift (CWV CLS)	Unexpected visual instability; **unitless score**	Individual layout-shift value is based on affected viewport area and movement distance. Google's current CWV aggregation takes the **largest session window** of unexpected shifts, with shifts separated by less than 1 s and a maximum 5 s window; shifts associated with recent user input are excluded from the unexpected-shift metric.	Layout Instability API entries followed by session-window aggregation. **Derived.**	Both, but full-lifecycle field measurement is particularly important
Interaction to Next Paint (INP)	End-to-end responsiveness of user interactions; time, ms	Event Timing groups events into interactions using `interactionId`. An interaction's latency spans from user input through input delay and event processing to the next presented frame; INP selects a near-worst interaction over the page lifetime, with high-interaction pages dropping one extreme interaction per 50 interactions. Relevant interaction classes include click/tap/keyboard interaction; scroll and hover are not INP interactions.	Performance Event Timing entries + event grouping + page-level outlier-selection algorithm. **Derived.**	Field; also local/lab when realistic interactions are executed

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
First Input Delay (FID)	Delay before the browser begins handling the first qualifying interaction; time, ms	Essentially `processingStart - startTime` for the first qualifying input. It ends when event processing begins—not when application code finishes or a resulting frame appears.	Event Timing/ first-input primitive. **Derived difference of timestamps**, but exposed as a defined metric.	Primarily field because an input is required
Total Blocking Time (TBT)	Laboratory estimate of main-thread unavailability that could delay input; time, ms	Lighthouse sums the **blocking portion** of long tasks between FCP and TTI. For each task longer than 50 ms, blocking contribution is the portion beyond 50 ms.	Derived from long-task/trace timing plus FCP/ TTI boundaries. **Derived.**	Laboratory/ synthetic

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
Speed Index (SI)	Visual loading progression in the recorded viewport; time-like score, commonly displayed in seconds	Lighthouse records page-load video, calculates visual completeness across frames, and uses the Speedline implementation based on WebPageTest's original Speed Index principles. Conceptually, time spent visually incomplete contributes to the score.	Screenshot/ video analysis. **Algorithmically derived.**	Laboratory/ synthetic
Time to Interactive (TTI)	Historical estimate of when a page becomes consistently interactive; time, s/ms	Lighthouse's historical metric combined useful-content appearance with main-thread/network quietness and a response-readiness heuristic rather than observing a specific user interaction.	Trace/network heuristic. **Derived.**	Laboratory; difficult and discouraged in field

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
Time to First Byte (TTFB)	Client-observed latency until response begins; time, ms	For a navigation, conceptually the interval from navigation timing origin/start to `responseStart`. It includes everything on the observed request path before the browser receives the response start—not merely server application execution.	Navigation Timing timestamps; **derived subtraction of direct timestamps.**	Both
Navigation / Resource Timing phases	Browser-visible network/ request lifecycle: DNS, connection, request, response, etc.; timestamps and durations, ms	`PerformanceNavigationTiming` and `PerformanceResourceTiming` expose high-resolution timestamps for defined milestones; phase duration is usually computed by subtraction. Reused connections, cache paths and other skipped phases can make timestamps equal or zero.	Standard Web Performance APIs. Timestamps are **direct browser observations**; phase durations are **derived**.	Both

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
`transferSize`, `encodedBodySize`, `decodedBodySize`	Browser-reported resource-size quantities; bytes	Resource Timing distinguishes encoded response body, decoded body, and transfer size. For `transferSize`, the current specification uses **0 for local-cache retrieval, 300 for a validated cache response, otherwise encoded body size plus a fixed 300-byte header allowance**.	Resource Timing counters. **Browser-derived byte quantities.**	Both
Server Timing	Server-defined processing/ routing/cache/ application measurements; typically durations in server-defined metrics	`Server-Timing` allows a server/proxy to communicate named performance measurements concerning the request-response cycle, surfaced as `PerformanceServerTiming`.	**Server-reported**, then browser-exposed—not independently inferred by browser.	Both, where server cooperates

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
JavaScript / scripting CPU time	Browser CPU spent in selected JavaScript-related trace activities; time, ms	Chrome traces expose activities such as Evaluate Script, Function Call, event callbacks and GC, which DevTools groups under Scripting. A total “JS execution time” must define which events and measurement window are included.	Chromium trace/CDP event aggregation. Individual events are **observed**; category totals are **derived**.	Primarily lab/ instrumented runs
Style / layout / rendering / painting time	CPU/render-pipeline work; time, ms	DevTools distinguishes trace events such as Recalculate Style, Layout, Paint and Composite Layers. Totals require summing events over an explicit interval.	Chromium trace/CDP. **Derived aggregate from observed trace intervals.**	Primarily lab

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
Long Tasks / Long Animation Frames	Potential main-thread blocking and long-frame responsiveness problems; time, ms	Long Tasks exposes tasks around the 50-ms threshold and duration/attribution information. The newer Long Animation Frames API provides a frame-oriented diagnostic designed to reveal lengthy animation-frame work and richer script/rendering attribution.	Performance entries produced by browser; **direct thresholded observations**, with further aggregates derived.	Both in browsers exposing APIs; particularly useful diagnostically in lab/RUM
Memory usage	Resource footprint in bytes, but scope must be defined	WICG `measureUserAgentSpecificMemory()` estimates memory allocated by a web application including iframes/workers, with implementation-specific breakdown and cross-origin-isolation requirements. CDP `Runtime.getHeapUsage` reports JavaScript heap for the corresponding isolate. `performance.memory` is non-standard JS-heap data.	Estimate or implementation-specific heap counters. **Not one common metric.**	Controlled lab and, for supported page API, production/RUM

Metric / signal	Intended construct and unit	Technical semantics and boundaries	Mechanism; direct vs. derived	Lab / field
DOM characteristics	Structural complexity/ state: node, document, listener counts; counts or derived structural indices	CDP `Memory.getDOMCounters` returns current counts for documents, nodes and JavaScript event listeners. A custom in-page DOM count is possible but its tree/ frame/shadow scope must be separately specified.	Snapshot counters. **Direct counts**; any "DOM complexity score" is derived.	Both conceptually; CDP measurement primarily synthetic/ instrumented

Two further qualifications follow directly from this matrix. First, **measurement units do not imply construct equivalence**: LCP, INP, FCP, TBT, SI and TTFB may all ultimately be represented in milliseconds, yet they measure different phenomena. Second, a value may be numerically derived from browser timestamps while still being conceptually user-centric: INP and LCP are examples of metrics whose intended constructs are outcomes even though their final values require algorithms over lower-level entries. [27]

Corrected taxonomy and applicability

A useful taxonomy for the proposed profiler separates **outcomes**, **progress indicators**, **causal/diagnostic observations**, and **resource-state measurements** instead of placing every quantity under "loading metrics."

Measurement family	Appropriate members	What a profiler can legitimately say	What it should not imply
Loading outcome / prominent-content visibility	LCP	"Time at which the browser's LCP algorithm identified the final largest qualifying content candidate under this run's lifecycle."	"Time when all content loaded," "time when the page became useful," or "time when the user perceived completion." LCP is explicitly heuristic. [28]
Initial visual appearance	FCP	"First qualifying contentful paint after navigation."	"Main content loaded." [29]

Measurement family	Appropriate members	What a profiler can legitimately say	What it should not imply
Responsiveness outcome	INP	“Near-worst latency of the qualifying interactions actually exercised/observed during the measured page lifecycle.”	“General interactivity” without specifying workload; scroll smoothness; response to interactions that never occurred. [30]
Responsiveness diagnostics	TBT, Long Tasks, Long Animation Frames, trace tasks	“Potential main-thread blocking under this controlled workload.”	“Measured INP” or “actual user input delay” when no interaction was executed. [31]
Visual stability	CWV CLS; underlying LayoutShift entries	“Unexpected layout instability according to the specified Layout Instability/CWV aggregation.”	General design quality, animation quality, or complete visual UX. [32]
Visual loading progression	Speed Index, screenshot-based completeness curves	“Rate at which the recorded viewport visually approached its reference/final state.”	Semantic importance or functional readiness. [33]
Client-observed request/network behavior	Navigation/Resource Timing, TTFB, DNS/ connect/TLS/request/ response phases, encoded/decoded body size	“Browser-observed request and response timing/size according to W3C timing semantics.”	Server execution time, packet-level traffic, or exact cross-origin waterfall data. [34]
Server-attributed behavior	`Server-Timing`	“Server-supplied named timing measurements for this request.”	Independently browser-verified internal server duration. [35]
Main-thread/ browser execution diagnostics	JavaScript evaluation/ calls, GC, style recalculation, layout, paint, compositing, task time	“Chromium trace activity within explicitly defined trace categories and intervals.”	Portable Web Platform metrics unless an interoperable specification exists. [36]
Resource footprint/state	User-agent-specific memory estimates, JS heap, DOM nodes/ documents/listeners	“Memory or structural state with stated API scope and observation time.”	A universal page-memory metric or direct user-experience score. [37]

Measurement family	Appropriate members	What a profiler can legitimately say	What it should not imply
Historical/retired metrics	FID, TTI; older FMP/ First CPU Idle metrics if encountered	“Legacy measurement reproduced for historical/tool-comparison reasons.”	Current Core Web Vital or preferred modern UX outcome. [38]

This taxonomy also corrects the often-overused **lab-versus-field binary**. An underlying browser API can be usable in both environments while the *interpretation of the derived metric* remains field-centric. LCP and CLS illustrate this: both can be captured synthetically, yet CWV quality assessment is defined around real-user distributions. INP can technically be measured during a controlled DevTools session or automated interaction sequence, but the value is conditional on the interactions that occurred. Google therefore describes CWV as fundamentally field metrics while explicitly allowing local measurements as debugging evidence. [39]

CrUX is not simply “the same metric measured more times.” It is a **real-user aggregate dataset drawn from eligible Chrome users and sufficiently popular/public origins and URLs**, with privacy filtering and normalization. Its page/origin values therefore represent a population statistic, not a replay of the conditions of a Lighthouse run. A profiler comparing its synthetic LCP of 1.8 s with a CrUX p75 LCP of 2.6 s is comparing a controlled observation with a distribution over heterogeneous real devices, networks, cache states, page states and users. [40]

This distinction becomes especially important for responsiveness. A Lighthouse navigation audit has no genuine sequence of real user interactions and consequently cannot produce a representative INP; **TBT may diagnose main-thread conditions that often contribute to poor INP, but it is explicitly not INP**. Conversely, DevTools can display local INP as the developer interacts with a page because interactions are actually occurring. [39]

There is also no defensible basis for treating the network and CPU families as independent causal dimensions. A slow resource can postpone script discovery; JavaScript can trigger further requests; long main-thread tasks can delay rendering after bytes have arrived; cached resources can eliminate network transfer without eliminating decode/layout/paint cost. A profiler should therefore describe these as **temporally related observations**, not assume that summing them yields a causal decomposition of page-load time. Navigation Timing, Resource Timing and Chrome's rendering traces expose different portions of this dependency structure. [41]

Terminology, semantic non-equivalence, and construct validity

A central finding is that **identical names across tools do not guarantee identical reported objects**, even where they share a common lower-level specification.

Common label	Why comparison is easily misinterpreted
“LCP”	The LCP API produces a sequence of candidate entries during one document lifecycle; a Web Vitals library derives a final page value; Lighthouse reports the value from one controlled audit; DevTools may show a local value for the currently exercised page; CrUX reports a real-user distribution/percentile. The underlying concept is related, but the statistical unit, workload, lifecycle, browser configuration and population are not the same. [42]
“CLS”	Layout Instability exposes individual shift entries. Google's current Core Web Vital CLS applies session-window aggregation, whereas terminology in the incubated Layout Instability specification also discusses document-level cumulative scores. A naïve sum of all `layout-shift` values is therefore not necessarily current CWV CLS. [32]
“INP” / “interaction latency”	Event Timing entries are primitives; INP is an algorithm over grouped interactions. A local DevTools INP reflects the interactions just exercised, while CrUX's value describes a field population. Navigation-only Lighthouse cannot produce an equivalent INP and instead recommends TBT diagnostically. [43]
“TTFB” / “server response time”	Browser TTFB is client-observed latency to response start. It does not separate origin application time from DNS, connection, TLS, cache/CDN, proxy or network effects. Server Timing is needed for explicit server-side phase attribution. [44]
“Transferred bytes”	Resource Timing `transferSize` deliberately includes a standardized 300-byte approximation and cache sentinel behavior; it is not exact TCP/QUIC/HTTP packet bytes. A DevTools Network panel byte figure, HAR field, Resource Timing property and packet capture must therefore be definition-matched before comparison. [45]
“JavaScript execution time”	This is usually an aggregate over selected Chromium trace events rather than a W3C metric. Whether compilation, GC, microtasks, callbacks, extension execution or worker execution is included depends on instrumentation and aggregation definitions. Chrome has changed trace presentation/categories over time. [46]
“Rendering time”	Could mean style recalculation + layout, paint only, paint + compositing, all rendering-category trace events, main-thread rendering only, or end-to-end frame presentation. No unqualified value should be compared across profilers. [47]
“Page memory”	`performance.memory`, `measureUserAgentSpecificMemory()`, CDP isolate heap, heap snapshots and browser-process memory have different scopes. They are not aliases for one measurement. [48]
“DOM size”	CDP's renderer DOM counters, a JavaScript traversal of the current document, an accessibility tree, and a diagnostic tool's DOM-complexity score are different objects. The scope must be stated. [26]
“Lighthouse performance”	Lighthouse's **performance score** is itself a normalized weighted composite, not another raw timing metric. Its current documented scoring model combines FCP, Speed Index, LCP, TBT and CLS, converting raw values through scoring distributions before weighting them. It therefore cannot be interpreted as a physical measurement scale. [49]

Construct validity

The metrics also differ in **what latent aspect of user experience they plausibly operationalize**.

LCP operationalizes a notion of prominent content visibility by selecting the largest eligible viewport content. That is a defensible proxy for loading experience, but it does not establish that the selected content is the user's task-relevant content. The LCP specification explicitly describes the mechanism as heuristic and discusses error cases, which is unusual but valuable evidence against treating LCP as a literal measurement of “when the page was useful.” [50]

FCP is still narrower: it provides evidence that the page has begun to render meaningful DOM content, but a page can produce an excellent FCP by rapidly painting a header or loading indicator while its principal content remains absent. LCP was introduced precisely because milestones such as FP/FCP do not reliably correspond to the point at which the page's main content appears. [51]

INP has stronger face validity for responsiveness than FID because it spans the full latency to presentation and considers interactions across the page lifecycle rather than merely the input delay of the first event. Yet INP is still conditional on the **interaction distribution**. A page containing an extremely slow rarely used control can receive a good run-level INP if that control is never exercised; another test sequence may expose it. Conversely, a workload that systematically triggers an expensive feature will measure a different operational behavior. [52]

CLS measures a different phenomenon altogether. A layout can remain geometrically stable while being visually confusing or slow, and a layout shift can be deliberately caused by user input and therefore excluded from CWV CLS. A low CLS must consequently be read as evidence of low *unexpected geometric instability according to the algorithm*, not evidence of an otherwise good UI. [32]

Speed Index is particularly illustrative of the difference between measurement and perception. It mathematically rewards rapid visual completion of the viewport, but does not know which pixels the user attends to. Kelton et al.'s gaze work found poor correspondence between traditional page-load metrics and user-perceived completion and demonstrated that user attention concentrates on particular regions. Gao, Dey and Ahammad likewise studied perceived above-the-fold speed through large-scale crowdsourcing, reflecting the broader observation that visual completion and perceived completion are related but not identical constructs. [53]

The same issue is even stronger for **interactivity**. Netravali, Nathan, Mickens and Balakrishnan defined Ready Index around both viewport visibility and availability of the JavaScript/DOM state supporting interactive elements; their Vesper experiments found that prior page-load metrics could under- or overestimate this functional readiness by 24–64%. Thus two pages with similar Speed Index—or even similar LCP—can have very different functional readiness. [54]

Mobile behavior further weakens a fixed-viewport interpretation of loading quality. Kelton et al.'s 2020 MobileHCI study explicitly found that conventional user-PLT methods focused on the initial viewport were ill suited to mobile behavior because users can begin scrolling while content is still loading; their model incorporated activity across multiple viewports. [55]

It follows that **two sites can have identical values for one metric and markedly different UX without any measurement error occurring**. Two pages with LCP = 2.0 s may differ radically in INP, CLS, content relevance, animation fluidity, post-load network activity or memory pressure. Even two pages matching on LCP, CLS and INP can differ in accessibility, application correctness, task completion, scroll performance, energy usage and content quality, because none of those are constructs defined by the three CWV algorithms. Google itself describes CWV as three *facets* of experience rather than a complete UX ontology. [56]

Contradictory evidence and unresolved debates

The most important apparent contradiction is between the **usefulness of Core Web Vitals as standardized engineering signals** and their **limited validity as holistic predictors of subjective QoE**. Google describes CWV as real-world signals reflecting critical user-centric outcomes and recommends their broad adoption. [56] Yet Wehner, Amir, Seufert, Schatz and Hoßfeld's 2022 QoMEX experiments found lower predictive value for CWV than expected, and the expanded 2023 study by Wehner, Seufert, Schatz and Hoßfeld again concluded that conventional loading-time effects remained highly important to subjective web QoE. These findings do not show that CWV are invalid; rather, they reject an interpretation of CWV as a sufficient model of total subjective experience. [57]

There is no genuine standards contradiction in LCP's current status, but there is an instructive tension: LCP is a **stable Core Web Vital while the underlying W3C specification remains a Working Draft and explicitly labels the algorithm heuristic and potentially error-prone**. “Stable Web Vital” is therefore a Google metric-lifecycle designation, not the same thing as W3C Recommendation maturity or proof that the construct perfectly captures perception. [58]

CLS presents a more concrete terminology problem. The Layout Instability work defines the underlying shift primitive and document-level concepts, but Google's current CWV CLS uses the maximum session-window aggregation introduced to prevent indefinitely long-lived pages from accumulating unbounded scores. Anyone implementing “CLS = sum of all layout-shift scores” from the primitive API can consequently produce a different number from current CWV CLS while believing the same metric name has been implemented. [32]

The replacement of FID by INP demonstrates that metric validity is not static. FID was a Core Web Vital but was judged too narrow because it covered only first-interaction input delay. INP broadened the construct to interaction latency over page lifetime, including processing and presentation components. A profiler whose schema hard-codes “Core Web Vitals = LCP, CLS, FID” would already be conceptually obsolete. [59]

TTI demonstrates a related issue on the synthetic side. Lighthouse retired it because sensitivity to outlier network requests and long tasks generated high variability. TBT remains useful because it captures blocking behavior during load, but Google explicitly treats it as a laboratory diagnostic/proxy rather than as the user outcome itself. The history cautions against treating any current derived heuristic as permanent merely because a major tool reports it. [60]

There is also an unresolved **scope problem for modern long-lived applications**. Traditional Navigation Timing begins with a document navigation, whereas single-page applications may execute substantial route transitions without a new hard navigation. CrUX documentation has historically noted the difficulty of attributing SPA experiences to subsequent route URLs because platform navigation boundaries did not

expose those transitions in the same manner as full navigations. Work on soft-navigation measurement is evolving, so a 2026 profiler should not silently equate “page lifecycle” with “user-visible route lifecycle.” [61]

Cross-origin content creates another unresolved practical mismatch between what Chrome internally knows and what page JavaScript may be permitted to observe. Resource Timing deliberately suppresses sensitive cross-origin information without the necessary opt-in, and performance APIs impose frame/origin boundaries for privacy. Consequently, a browser-internal trace, CDP client, page-injected PerformanceObserver, and CrUX aggregation may have different observability even when they nominally report the same class of phenomenon. [62]

Memory remains even less settled. The WICG memory proposal explicitly exists because traditional JS-heap reporting can both overestimate and underestimate a page's actual memory footprint. It permits estimated results and implementation-specific attribution. This means there is presently no standards-backed scalar that supports the unconditional statement “this URL uses X MB of browser memory.” A defensible statement must name the measurement scope. [48]

Finally, the literature does not support a single universal objective function for web loading. Gaze-based work emphasizes attention; Vesper emphasizes visual plus functional readiness; MobileHCI work incorporates scrolling behavior; QoMEX work questions how strongly CWV predict overall QoE; Google's Web Vitals intentionally favors a small, operationally tractable set of broadly applicable signals. These are better understood as **different operationalizations of partly overlapping constructs** than as competing attempts to measure one self-evident ground-truth quantity called “web performance.” [63]

Implications for a custom browser profiler

The following separates propositions strongly grounded in specifications/tool documentation from design inferences that follow from those findings but are not themselves mandated by the literature.

Implication	Evidential status	Rationale
Store raw performance entries or trace primitives separately from derived metrics.	**Literature/ specification-supported.**	LCP, CLS, INP and TBT are algorithms over lower-level paint, shift, event and task observations. Preserving primitives allows algorithm revisions to be audited rather than confusing collection with interpretation. [64]
Version every derived metric definition.	**Strong researcher inference from documented metric evolution.**	CLS aggregation changed; FID was replaced by INP; TTI was retired. A value named only `"cls": 0.12` is insufficient provenance for longitudinal research. [65]

Implication	Evidential status	Rationale
Represent `metric_name`, `metric_definition_version`, browser/ version, collection mechanism, lifecycle boundary and workload as separate metadata.	**Researcher inference.**	The evidence shows that same-name values can differ because of lifecycle, tool and population rather than arithmetic error. This schema would make those conditions explicit. [66]
Do not expose TBT as "interactivity" without the word laboratory/diagnostic.	**Literature-supported.**	Google explicitly distinguishes TBT from field INP and treats it as a lab diagnostic/proxy. [67]
Do not expose browser TTFB as "server processing time."	**Specification-supported.**	Browser timing cannot decompose server routing/ application time; `Server-Timing` exists specifically to communicate such internal metrics. [68]
Expose Resource Timing byte properties by their exact names rather than a generic `network_bytes`.	**Specification-supported.**	`transferSize`, `encodedBodySize` and `decodedBodySize` have materially different semantics, and `transferSize` is not exact wire bytes. [69]
Separate standards-based browser APIs from Chromium-only/CDP measurements in the API schema.	**Literature/ specification-supported; schema separation is inference.**	Navigation, Resource, Paint and Event Timing have Web Performance specifications; Chrome trace categories and CDP DOM/heap counters are browser-specific instrumentation. [70]
Treat JavaScript, style, layout and paint totals as interval-defined diagnostic aggregates, not universal metric names.	**Tool-documentation-supported.**	DevTools obtains these concepts from trace event categories. The aggregation window and included event classes must therefore be part of the definition. [47]
Label memory values with scope, e.g. `ua_specific_page_memory_estimate`, `v8_isolate_heap_used`, rather than `memory_usage`.	**Strong researcher inference from API semantics.**	The available memory APIs explicitly measure different scopes. [48]

Implication	Evidential status	Rationale
Treat DOM node/document/listener counts as explanatory state variables.	**Literature-supported classification; causal use is inference.**	CDP exposes them as counters; no cited standard defines them as user-centric performance outcomes. [26]
For lab INP, record the exact interaction script and state transitions alongside the value.	**Strong researcher inference.**	INP is computed only from interactions that occur; without workload provenance, two lab INP values do not necessarily measure comparable behavior. [30]
Do not compare a synthetic run directly with a CrUX percentile as if they were repeated samples from one experiment.	**Literature-supported.**	CrUX is an aggregate real-user population subject to eligibility/privacy rules; laboratory runs operate under controlled device/network/browser/workload conditions. [71]
Preserve browser and profiler versions with traces.	**Strong researcher inference from implementation evolution.**	Chrome's trace presentation/categories and Lighthouse metric set have changed, including removal of TTI and changes to trace-category presentation. [72]
Avoid collapsing heterogeneous metrics into an unexplained scalar "performance score."	**Literature-supported concern; API-design consequence is inference.**	Lighthouse's own score demonstrates that any such scalar requires explicit normalization distributions and weights; it is not a direct physical quantity. [49]
Where possible, report uncertainty/distribution across repeated controlled runs rather than only one number.	**Researcher inference strongly motivated by documented variability.**	TTI's retirement because of high variability and the general lab/field distinction show that single observations can be unstable and environment-dependent. [73]

A particularly useful architectural implication is to distinguish **observation**, **normalization**, and **interpretation**. Observation would retain native timestamps, entries, counters and trace events. Normalization would transform units and identifiers without changing metric semantics. Interpretation would implement named/versioned algorithms such as CWV CLS session windows, INP interaction

selection, TBT blocking sums, Speed Index visual progression, or custom aggregate CPU categories. This three-layer split is a **researcher inference**, but it follows directly from the fact that the standards expose primitives while several widely used metrics are higher-level algorithms over those primitives. [74]

“Normalized” should therefore **not** mean making unlike measurements appear comparable. Converting every duration to milliseconds is harmless unit normalization; transforming a CrUX p75 LCP, one Lighthouse LCP, one local DevTools LCP and a raw `LargestContentfulPaint` candidate into a common field called `lcp_ms` without provenance would erase essential semantics. [75]

For similar reasons, the profiler should be careful with causal language. A trace showing 600 ms of script activity before LCP establishes temporal work observed before LCP; it does not by itself prove that eliminating all 600 ms would improve LCP by 600 ms. Browser pipelines contain dependencies, parallelism and idle/blocked intervals. The evidence surveyed here supports **measurement and association**, while causal attribution requires a separate experimental design that is outside the requested measurement-foundations scope. [76]

Priority primary reading and open questions

The following are the primary specifications and original studies that are most worth preserving locally and examining manually. A DOI is included only where a genuine DOI/publication identifier could be verified; W3C/WICG specifications generally do not have DOIs.

Work to examine	Verified bibliographic details	DOI / publication identifier
Paint Timing	W3C Web Performance Working Group, *Paint Timing*, W3C Working Draft, **24 March 2026**. Fundamental source for FP/FCP semantics. [77]	No DOI
Largest Contentful Paint	W3C Web Performance Working Group, *Largest Contentful Paint*, W3C Working Draft, **26 August 2026**. Current normative source for LCP entries and processing semantics. [78]	No DOI
Event Timing API	W3C Web Performance Working Group, *Event Timing API*, W3C Working Draft, **19 March 2026**. Underlying event/interaction primitives for INP and first-input measurements. [79]	No DOI
Layout Instability API	Web Platform Incubator Community Group, *Layout Instability API*, Community Group Report/ current incubated specification; current report surfaced as **17 December 2025**. Important because the primitive API and CWV CLS aggregation must be distinguished. [80]	No DOI
Navigation Timing Level 2	W3C Web Performance Working Group, *Navigation Timing Level 2*, W3C Working Draft, **25 February 2026**. [81]	No DOI

Work to examine	Verified bibliographic details	DOI / publication identifier
Resource Timing	W3C Web Performance Working Group, *Resource Timing*, Candidate Recommendation Draft, **20 April 2026**. Essential for request-phase and byte semantics. [82]	No DOI
High Resolution Time	W3C Web Performance Working Group, *High Resolution Time*, current Level 3 Working Draft, **24 March 2026**. Defines time origins, monotonic clocks and allowed precision reduction. [10]	No DOI
Server Timing	W3C Web Performance Working Group, *Server Timing*, W3C Working Draft, **7 April 2026**. [22]	No DOI
Long Tasks API	W3C Web Performance Working Group, *Long Tasks API*, current 2026 working-draft material. Primary source for long-task thresholding and attribution semantics. [83]	No DOI
Long Animation Frames API	W3C Web Performance Working Group, *Long Animation Frames API*, current 2026 working-draft material. Particularly relevant for richer script/rendering responsiveness diagnostics. [84]	No DOI
Performance Measure Memory proposal	WICG, *Performance Measure Memory /* `measureUserAgentSpecificMemory()`, incubated specification/proposal. Defines page-attributed estimated memory and explains why `performance.memory` has incorrect scope for many use cases. [85]	No DOI
Kelton et al., Improving User Perceived Page Load Times Using Gaze	Conor Kelton, Jihoon Ryoo, Aruna Balasubramanian, Samir R. Das. *14th USENIX Symposium on Networked Systems Design and Implementation (NSDI '17)*, 2017, pp. 545–559. Original user-perception/gaze study. [86]	`10.5555/3154630.3154675` in the ACM bibliographic record
Gao, Dey & Ahammad, Perceived Performance of Top Retail Webpages in the Wild	Q. Gao, P. Dey, P. Ahammad. “Perceived Performance of Top Retail Webpages In the Wild: Insights from Large-scale Crowdsourcing of Above-the-Fold QoE.” *Internet-QoE '17: Workshop on QoE-based Analysis and Management of Data Communication Networks*, 2017, pp. 13–18. [87]	`10.1145/3098603.3098606`

Work to examine	Verified bibliographic details	DOI / publication identifier
Netravali et al., Vesper	Ravi Netravali, Vikram Nathan, James Mickens, Hari Balakrishnan. “Vesper: Measuring Time-to-Interactivity for Web Pages.” *15th USENIX Symposium on Networked Systems Design and Implementation (NSDI '18)*, 2018, pp. 217–231. ISBN 978-1-939133-01-4. [54]	No DOI identified in the original USENIX publication
Hoßfeld, Metzger & Rossi, Speed Index	Tobias Hoßfeld, Florian Metzger, Dario Rossi. “Speed Index: Relating the Industrial Standard for User Perceived Web Performance to Web QoE.” *2018 Tenth International Conference on Quality of Multimedia Experience (QoMEX)*, pp. 1–6. [88]	`10.1109/QoMEX.2018.8463430`
Kelton et al., Modeling User-Centered Page Load Time for Smartphones	Conor Kelton, Jihoon Ryoo, Aruna Balasubramanian, Xiaojun Bi, Samir R. Das. *22nd International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI 2020)*, 2020, pp. 1–12 / article 3403565. [89]	`10.1145/3379503.3403565`
Wehner et al., A Vital Improvement?	Nikolas Wehner, Monisha Amir, Michael Seufert, Raimund Schatz, Tobias Hoßfeld. “A Vital Improvement? Relating Google's Core Web Vitals to Actual Web QoE.” *2022 14th International Conference on Quality of Multimedia Experience (QoMEX)*, pp. 1–6. [90]	`10.1109/QoMEX55416.2022.9900881`
Wehner et al., Do you agree?	Nikolas Wehner, Michael Seufert, Raimund Schatz, Tobias Hoßfeld. “Do you agree? Contrasting Google's Core Web Vitals and the impact of cookie consent banners with actual web QoE.” *Quality and User Experience*, vol. 8, article 5, 2023. [91]	`10.1007/s41233-023-00058-3`
van Riet, Malavolta & Ghaleb, Optimize along the way	Jasper van Riet, Ivano Malavolta, Taher A. Ghaleb. “Optimize along the way: An industrial case study on web performance.” *Journal of Systems and Software*, vol. 198, article 111593, 2023, pp. 1–24. [92]	`10.1016/j.jss.2022.111593`

Several questions remain open enough to warrant focused investigation before later methodological decisions are made.

Candidate unanswered question	Why it matters
What should constitute one measurement lifecycle for a single-page application?	Hard-navigation APIs have clear document boundaries, but route transitions, long-lived CLS/INP accumulation and emerging soft-navigation semantics complicate what a profiler should regard as one “page.” CrUX itself has documented SPA-attribution limitations. [61]
What exact interaction workload is required for a reproducible laboratory INP?	INP depends on which interactions occur. A synthetic value cannot be comparable across sites or runs unless interaction selection, timing and application state are specified. [30]
How closely can browser-internal CWV calculations, page-level Web Performance APIs and the `web-vitals`/CrUX implementations be reproduced byte-for-byte or event-for-event?	The same underlying concept can have lifecycle, iframe, privacy, finalization and aggregation differences. [93]
How should cross-origin iframe contributions be handled?	Browser internals can have information unavailable to top-level JavaScript, while security/privacy restrictions constrain what a custom page-level observer can inspect. [94]
Which Chromium trace events constitute reproducible “JavaScript,” “layout,” “rendering,” and “painting” aggregates across browser versions?	These are useful diagnostics but not standardized metrics, and DevTools trace categorization evolves. [46]
What memory scope is scientifically most meaningful for a URL-level profiler?	JS isolate heap, page-attributed estimated memory, worker memory and total renderer/process memory answer different questions; the current platform offers no single universal scope. [48]
How should garbage collection and memory-state history be controlled when comparing memory measurements?	Memory is stateful rather than a one-way completion milestone, making observation point and workload history potentially as important as the numeric value. The available APIs already differ substantially in scope and estimation method. [95]
When is `transferSize` sufficient, and when is packet- or protocol-level capture necessary?	Resource Timing's intentionally approximate header accounting prevents its interpretation as exact network traffic; studies of compression, HTTP/2/3 framing, retransmission or packet loss may require a lower measurement layer. [69]

Candidate unanswered question	Why it matters
How stable are synthetic metrics under browser version, CPU throttling, cache state and network-control mechanism?	Lighthouse's retirement of TTI because of variability and the general lab/field discrepancy demonstrate that execution conditions can materially affect derived results. [73]
What perceptual weighting, if any, should supplement pixel-based visual progression?	Gaze and crowdsourced QoE research shows that different regions have different perceptual importance, while Speed Index operates on visual progression rather than semantic user attention. [96]
How should diagnostic measurements be linked to outcome metrics without asserting unsupported causality?	Network, main-thread and rendering events form dependent pipelines; temporal overlap or precedence alone does not identify causal contribution to LCP, CLS or INP. [97]
What should "equivalent measurement conditions" mean for cross-tool comparison?	At minimum, evidence here suggests matching browser/ version, viewport, device/CPU conditions, network shaping, cache state, navigation type, lifecycle duration, interaction workload, iframe policy and metric algorithm version; determining which of these are necessary for each metric remains an empirical question. [98]

Taken together, the specifications and empirical literature support a measurement foundation in which **LCP, INP and CLS are the current Core Web Vitals but not a complete theory of web performance; FCP, TBT and Speed Index remain useful but differently scoped measurements; FID and TTI are historical/ retired for current primary evaluation; request, CPU, memory and DOM measurements are principally explanatory diagnostics; and identical metric names are comparable only after their algorithm, lifecycle, environment, instrumentation source and statistical unit have been matched.** [99]

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